Computer Science Department

Senior Project Requirements and Design – Fall 2025

Title of the project:	"Smart Campus Navigation with Real-Time Positioning and Adaptive Accessibility Routing"
Team Members:	Aiym Abay, Madina Dairova, Zhansaya Zhassulanova, Aiymzhan Ashirbek, Aruzhan Ryssayeva
Project Advisor/Co-Advisors	Askar Boranbayev

Project Overview (20%)

Our project is about helping students, teachers, and visitors find their way around campus more easily. Right now, apps like Google Maps or 2GIS don't really work well indoors and don't have detailed campus information. Because of that, people often get lost trying to find classrooms, especially new students or anyone who needs accessible routes.

We plan to create a mobile and web app that gives real-time directions both inside and outside of campus buildings. The app will use GPS, Wi-Fi, and Bluetooth beacons to make the location more accurate and suggest routes based on what the user prefers — for example, avoiding stairs or picking the fastest way. It will also show live updates about things like events, construction, and crowded areas. Besides navigation, we want the app to have extra tools that are actually useful for students, like a GPA calculator, class planner, and discount system. The backend will use cloud services so the app works fast and can handle many users at once.

In short, our goal is to make campus life easier by building a simple, reliable, and inclusive app that brings navigation, accessibility, and student tools together in one place.

Requirements (30%)

Functional Requirements:

- GPS outdoor positioning (±3m) and Wi-Fi/beacon indoor positioning (≤2m) with seamless transitions
- Multi-criteria routing: shortest, accessible, and least crowded paths
- Real-time rerouting based on events, construction, and crowd density
- Fallback positioning using Wi-Fi fingerprinting and sensors (≤5m)
- ARCore/ARKit integration with real-time overlays and landmark recognition at 60fps
- 2D fallback for non-AR devices and WebXR web support
- Multi-category reviews with NLP sentiment analysis and anti-spam mechanisms
- Personalized recommendations for study spots, dining (with dietary preferences), and events
- ML-based crowd prediction with rule-based fallbacks

- External API integration with fraud prevention
- Student verification and loyalty points tied to engagement
- Secure payment processing
- GPA calculator with grade prediction
- Academic planner with degree tracking
- Google Calendar sync with smart scheduling
- Course system integration with manual fallback
- OAuth 2.0/JWT authentication with anonymous mode
- PostgreSQL (structured) and MongoDB (reviews)
- Offline-first with cached maps and pre-computed routes
- GDPR-compliant export/deletion and crowdsourced mapping
- Real-time metrics, behavior tracking, and sentiment trends
- Performance monitoring with alerts

Non-Functional Requirements:

- 2-second route calculation supporting 1,000+ concurrent users
- Microservices with Docker on cloud (AWS/Azure/Google Cloud)
- 99.5% uptime with automated failover
- End-to-end encryption with GDPR compliance
- Differential privacy and regular security audits
- Material Design/iOS Guidelines with multi-language support
- WCAG 2.1 Level AA with audio guidance
- 4.0/5.0+ user satisfaction target
- iOS 13+, Android 8.0+ (Flutter/React Native), and PWA
- Sentiment analysis >85% precision/recall
- Modular architecture with CI/CD pipeline
- Comprehensive documentation and installation guides

Domain Requirements:

- Room-level building maps with construction/event tracking
- Course registration and scheduling integration
- University-specific GPA and degree tracking
- Dining, library, event, and security system integration
- Campus transportation and payment system support
- University SSO with role-based access
- ADA-compliant routes with elevator/ramp identification
- Emergency exits, alerts, and evacuation guidance
- Offline functionality during emergencies
- WGS84 and IndoorGML standards with multi-floor support
- Peak usage handling during class transitions
- FERPA compliance with university IT policies
- Student discount verification and vendor management

Preliminary Design (20%)

Stakeholders

- Students (primary): navigation, accessibility, class planning, discounts.
- Faculty & Staff: schedule sync, indoor rerouting, event locations.
- Visitors: campus tours, step-free guidance.
- Accessibility Office: monitor and verify accessible routes.
- Facilities & Security: manage closures, safety alerts, evacuation paths.
- IT Services: authentication, data protection, system uptime.
- Vendors/Partners: student verification, discount offers.
- Project Team & Advisor: system design, development, and evaluation.

Use Cases and Sequence Diagrams

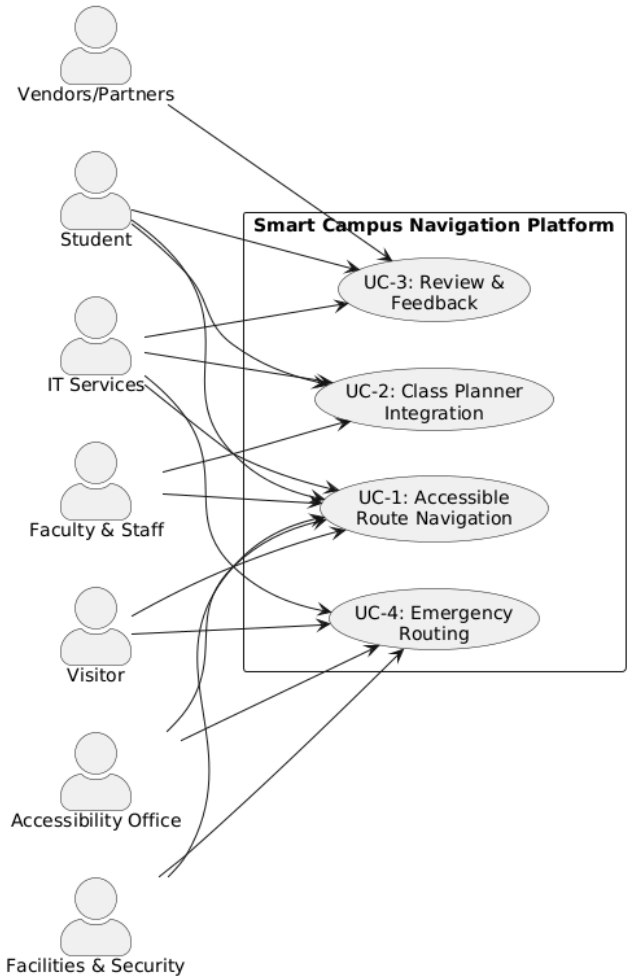

Figure 1. Use Case Diagram — Smart Campus Navigation

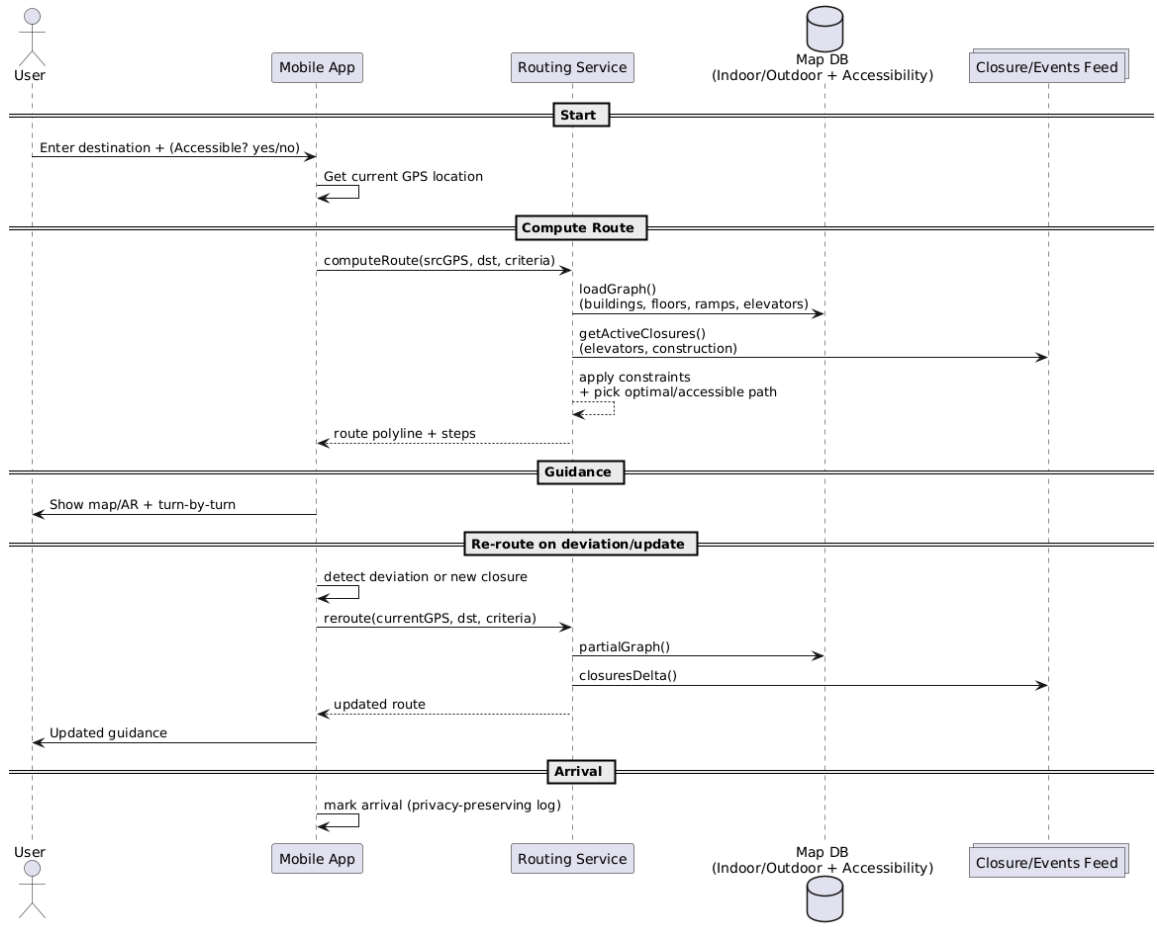

Figure 2. Sequence Diagram — Accessible Route Navigation

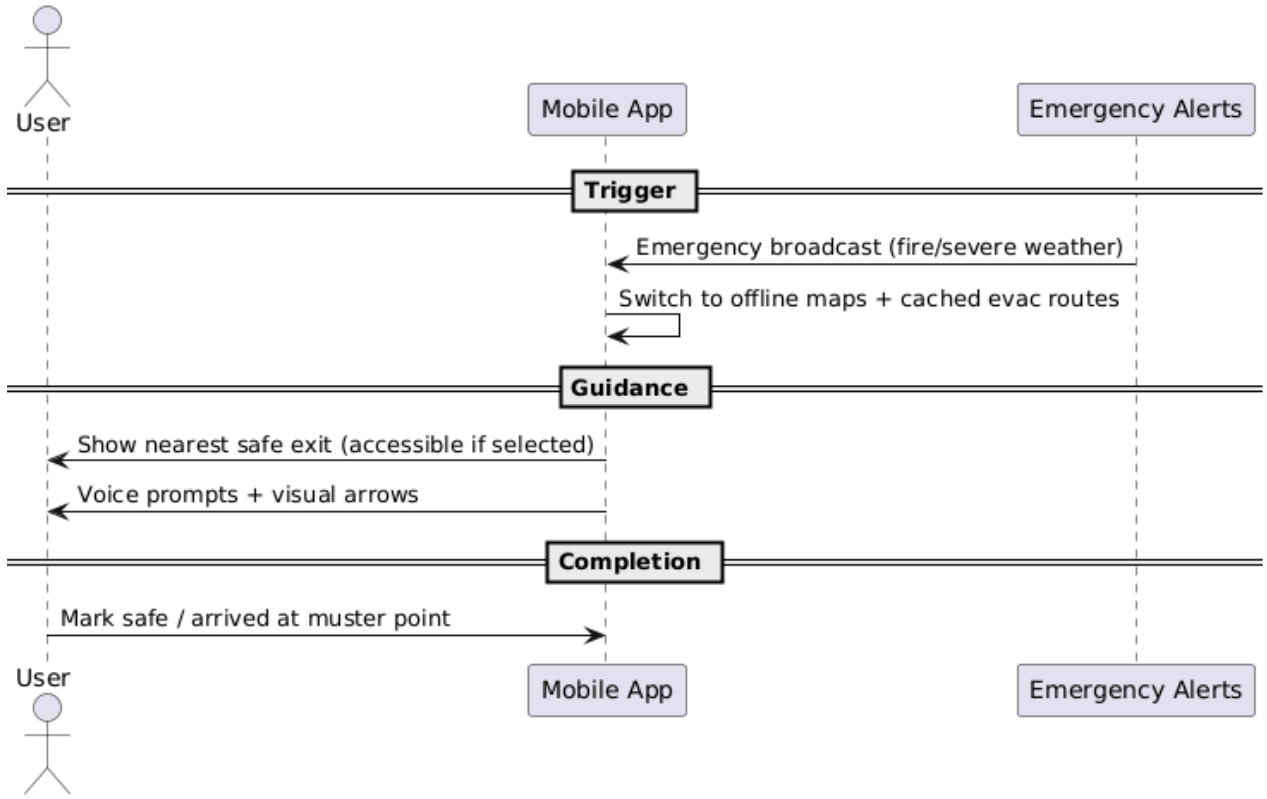

Figure 3. Sequence Diagram — Emergency Routing (Offline)

Architectural Overview

Smart Campus Navigation Platform integrates maps, schedules, and events for students and staff.

External systems: University SSO, Course API, Google Calendar, Vendor APIs.

Containers

- Mobile App: map, navigation, offline mode, reviews.
- API Gateway: single entry point, authentication.
- Routing Service: GPS-based route generation.
- Map Database: indoor/outdoor campus data, accessibility metadata.
- Planner Service: schedule and calendar integration.
- Review Service: user feedback and moderation.
- Infrastructure: PostgreSQL, Docker/Kubernetes, CI/CD pipeline.

Key Qualities

- Route generation under 2 seconds.
- Offline functionality during emergencies.
- 99.5% uptime.
- Accessible design (WCAG 2.1 AA).

- GDPR/FERPA-compliant data management.

High-level tasking plans (20%)

Project Management Infrastructure

The project will be managed using **Notion** and **GitHub Projects** as the primary tools.

Notion:

- The primary project management tool for task planning, shared documentation creation, and progress tracking across the development lifecycle.
- Kanban boards and Gantt charts integrated within Notion will be utilized for workflow organization and timeline visualization.

GitHub Projects will be employed specifically for technical task tracking, version control management, and code review coordination.

Project management will follow an **iterative and incremental approach**, divided into **two main phases** corresponding to the academic semesters.

Task Allocation and Member Responsibilities

Fall 2025 - Research, Foundation & Prototype Phase

- **System Architecture & Planning**
  Lead: **Madina Dairova**
  Design the backend architecture, database schemas, and API specifications to support hybrid positioning and data exchange.

- **AI & Data Pipeline Setup**
  Lead: **Aruzhan Ryssayeva**
  Conduct data analysis and model research for route optimization, crowd prediction, and sentiment analysis. Prepare datasets for ML model training and integrate early-stage analytics.

- **Mobile App Prototype Development**
  Leads: **Aiym Abay, Zhansaya Zhassulanova, Aiymzhan Ashirbek**
  Build the cross-platform mobile interface (Flutter/React Native) including authentication, map rendering, and preliminary route visualization.

- **UI/UX Design & Integration Testing**
  Leads: **Aiym Abay & Zhansaya Zhassulanova**
  Develop initial UI components and conduct early usability feedback sessions.

- **Progress Review & Prototype Demonstration**
  All members

Present architecture design and working prototype for feedback.

Spring 2026 - Integration, Testing, Optimization, and Deployment

- **Backend Development & Optimization**
  Lead: **Madina Dairova**
  Implement REST APIs, authentication, data synchronization, and cloud deployment. Ensure scalability, latency reduction, and security compliance.

- **AI, Machine Learning, and Data Analytics Integration**
  Lead: **Aruzhan Ryssayeva**
  Integrate trained models into the backend for recommendation systems, route prediction, and review sentiment analysis. Develop analytics dashboard to visualize app usage and user behavior.

- **Mobile App Finalization & Feature Integration**
  Leads: **Aiym Abay, Zhansaya Zhassulanova, Aiymzhan Ashirbek**
  Integrate discount and academic tools, AR navigation interface, and accessibility-aware routing. Conduct cross-platform optimization for iOS and Android.

- **Testing & Evaluation**
  All members
  Conduct unit, integration, performance, and accessibility testing. Collect user feedback and iterate.

- **Documentation & Final Presentation**
  All members
  Compile technical documentation, model evaluation reports, and final project presentation materials.

Conclusion (10%)

By the end of this semester, the team will manage to finish the main technical part of the Smart Campus Lifestyle Platform. Backend architecture and API specification within authentication, routing, reviews, and discounts will be finalized. Also, the database design and initial datasets will be set for further processing. A cross-platform mobile prototype will be developed, which includes login, visualization of the campus map, basic indoor/outdoor navigation, and testing panels of reviews and discounts. First elements of AR navigation will be added so we can demonstrate the concept of visual guidance in the next semester. Through these steps, by the end of Fall 2025, the project will have a working prototype and finished system design that will be used as a base for integration, testing, and further development of functionality in Spring 2026.